

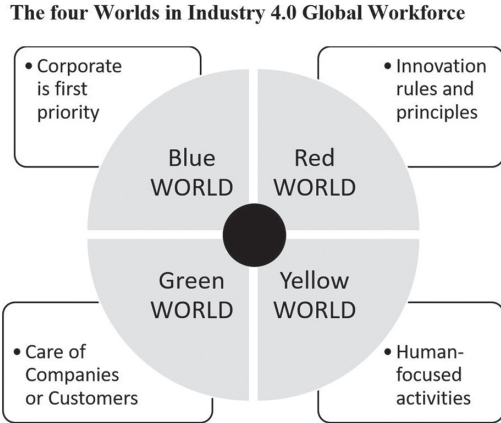


FIGURE 1.4 The four worlds in global workforce.

1.2.5.2 The Blue World Is All about Extreme Talent

The Blue World will be entirely talent-driven, which is essential for maintaining profit margins. Employees at all levels will strive to advance their professional growth and attempt to sharpen their skills through the use of cognition-enhancing methods, because talent will be so vital (Khang et al., 2022a,b).

1.2.5.3 The Green World Is All about the Company’s Caring

The Green World is characterized by a strong social conscience, environmental responsibility, an emphasis on diversity and human rights, and a knowledge that business has repercussions beyond the bottom line.

1.2.5.4 The Yellow World Is All about Humans Coming First

Here, humans come first. Compared to its Red and Blue counterparts, the Yellow World of work places a considerably higher emphasis on individuals. Locally and socially conscious businesses are the most prosperous.

1.2.6 BUILDING A CONNECTED WORKFORCE FOR INDUSTRY 4.0

It’s natural to believe that Industry 4.0 will eliminate chances for human labor in services, factories, mines, ports, and other settings emphasizing automation, AI, and digitalization. In reality, today’s occupations will change as Industry 4.0 technologies gain traction, and new job possibilities will be created by the new skills needed by the 4.0 workforce (Rani et al., 2021).

These new methods of working in the Fourth Industrial Revolution will result in workforces that are streamlined, nimble, and technology-augmented, making them safer, happier, healthier, and more suited to do their jobs, while boosting productivity for the corporations (Hahanov et al., 2022). This process may include the following.

1.2.6.1 Continue to Combat Knowledge Loss

The long-term threat to workforce management success may be knowledge erosion. Thus, it's crucial to include frequent ongoing training in your workforce management contract. Pay special attention to best practices and refresher modules that can assist in reskilling and upskilling.

1.2.6.2 Place a Priority on People, Processes, and Culture

There are so many powerful new capabilities that AI technology offers. But it is important not to be beguiled by technology at the expense of establishing a strong business foundation built on people, processes, and culture.

Ultimately, if the underlying processes don't change and the people and the culture of your organization are not ready for the technology, any automation and implementation will fail.

1.2.6.3 Strengthening the Employee Value Proposition

Employees get what in exchange for what they give to help retain talent in the most critical roles?

1.2.6.4 Rethinking the Talent Paradigm, Not Just Attracting and Retaining Employees

To create a more dynamic organization and place, you need focus on developing talent pipelines, upskilling current employees, and using shorter-term work arrangements.

1.2.6.5 Lead with Quality

It is imperative to identify and measure quality outcome indicators before, during, and after change implementation.

1.2.6.6 Set Direction with Comparative Benchmarking

Identify internal and external best performance practices and sustain continuous improvement through the process of benchmarking.

1.2.6.7 Elevate and Support through Coaching and Education

Technology is relatively easy to implement. The challenge is ensuring that the people who interact with the technology have the skills they need to get value out of it. Employees must not only be trained on how to operate new technology but on how to embed the technology in their daily work.

1.2.6.8 Conducting a Job Analysis to Understand How Work Is Currently Done

Use the data to determine how it could be done more efficiently in the age of AI.

1.3 CONCLUSION

The truth is that businesses need to start visualizing the future for themselves, because it is already here. The guidelines alter occasionally. What matters and how value is created in business are undergoing a fundamental upheaval. Talent has not

only taken the position of the land, capital, and raw materials as the main source of competitive advantage, but talent's objectives and expectations have also changed.

The exact design of workplaces in the future won't be determined solely by factors of technological and cultural growth. A change in how work is accomplished is also significant. The places where people work, the ways they communicate, and how business interactions and transactions are carried out and managed are all changing quickly (Hahanov et al., 2022).

Business executives should consider how they would respond if the world becomes more blue or green. Existing employees should consider their skill set rather than the position they currently hold. Even in near future, it will be impossible to forecast exactly what skills will be required. Therefore, people and organizations must be prepared to adapt to any of the futures we see.

A significant portion of advanced technical responsibilities will inevitably fall to the individual. In addition to being able to adapt to organizational change, they will need to be eager to learn new things throughout their lives, take on new responsibilities, and even rethink and retrain in the middle of their careers (Khang, Chowdhury, & Sharma, 2002).

This is of immense and vital importance to their workplace. In the age of AI technology, upskilling of the brownfield workforce by using hybrid strategies will support corporations in Industry 4.0 to achieve a level of greenfield workforce (Vrushank, Khang, & Rani, 2023). The framework proposed in this chapter will benefit researchers and corporations in overcoming the challenges and weaknesses of workforce management.

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2 Industry Revolution 4.0

Workforce Competency Models and Designs

*Alex Khang, Babasaheb Jadhav,
and Sphurti Birajdar*

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2.1 INTRODUCTION

Industry 4.0 describes the fundamental change in an industry that has occurred as a result of integrating modern technology. It suggests that the Fourth Industrial Revolution is about to begin. Industry 4.0 intends to make it possible for intelligent companies to produce individualized output using cleaner, more effective methods.

A centralized computer that oversees the responsibilities and operations of all the various units from the supply chain management (SCM) distribution can manage factories. The constant communication of information between all of the units is one essential function. Fast-moving technological advancements have turned the actual workplace into a virtual one, placing human resources (HR) at a juncture.

Robotics, virtual world, augmented reality, artificial intelligence (AI), and many other technologies have found their way into human resources functions, such as hiring, training, employee communications, data analytics, workforce directories, compensation and benefits management, retention strategies, and support programs, as well as other routine activities. A competency model is a collection of guidelines created by a human resources team that outlines the precise knowledge, abilities, and conduct needed for a person to do their job well (Rana et al., 2021).

Nowadays every HR professional is going through these three challenges: development of leadership skills, change management, and knowledge replenishment. To address these issues, HR with competencies has ushered in a new pattern. Searching for an appropriate person for the right job is only one aspect of the competency-mapping principle; in reality, the situation is considerably more complicated. It is aptly said that the organization's overall performance is gauged by its workforce's efficiency and productivity.

Digital existence is enhancing human potential while upending long-standing human activity. More than half of the world's population has access to ambient information and connectivity thanks to code-driven systems, which represent an interest in both the potential and dangers. In today's commercial context, AI technology is popular and has a significant impact on various organizational realms, particularly in Industry 4.0 (Khanh & Khang, 2021).

HR analytics can be used to retain top performers after using AI to identify those who are intending to leave or switch to the organization. To raise organizations to newer heights, AI thereby delivers transparency, cost-effectiveness, and reduced bias with data-driven HR decisions (Chowdhury and colleagues, 2022).

Industry 4.0 describes the tremendous change in industry that has occurred from the convergence of new technologies. The shortcomings of Industry 4.0 necessitate a transformation in the labor market, which requires a clearly qualified and professionally trained workforce, including the highest side of the competencies who are capable of flourishing in this novel environment.

The constant exchange of data among all the subsystems is one essential function. Industry 4.0 can improve the speed of decision-making, improve floor monitoring and control, optimize resource utilization, and demand forecasting. Industry 4.0 will have a significant impact on the work setting.

Employees must be equipped with a wide variety of competencies, because the changing nature of the workplace will alter jobs. Jobs that require a higher education will become more important in Industry 4.0, but automated processes will mostly replace the labor force. As a result, many practitioners and experts concur that one of the major barriers to implementing Industry 4.0 is the development of competencies for students and workers looking for jobs that demand higher education (Rani et al., 2023).

While the number of telecommuters has increased due to globalization and internationalization, supported by 4G and 5G data connections, AI has automated HR functions. (Manuti & de Palma, 2017).

By 2030, “the future of humans at work is questioned,” according to a PwC report titled “Workforce from the Future” (PwC, 2018). A new set of capabilities will be needed to handle the aforementioned trends, including globalization, virtual companies, HR automation, and AI.

HR has already moved into the virtual and mobile spaces, so it is necessary to identify and define futuristic capabilities to prepare the next generation for the occupations of the future with the age of AI and Internet of Things (IoT).

2.2 COMPETENCIES AND THE COMPETENCY MODEL

The concept of competency was developed and proposed by McClelland in the year 1973. Both practitioners and academics have researched the concept of competency extensively. McClelland contends that people should be examined for their competency rather than their intelligence.

According to Khang, Rani, and Shah (2024), competency is a quantifiable human ability needed for successful performance. The knowledge can be more sophisticated, abstract, and challenging to evaluate, or it might be concrete, particular, and easily measured. Knowledge is gained through instruction and practical application. A skill is an ability to carry out physical or mental tasks with the desired result.

A verified physical or intellectual ability to carry out a task successfully with a variety of possible outcomes is called an *ability*. A skill is frequently a confluence of several underlying capacities that give us the ability to learn and perform. These are typically time-consuming, challenging to learn, and heavily reliant on intrinsic ability.

For instance, some people are more naturally gifted with analytical thinking than others, and many people find it difficult to develop this skill as [Table 2.1](#).

From the above table, it is clear that the functions of human resource management (HRM) have revolutionized with the pace of the Industrial Revolution.

In Industry 1.0, mechanization had just started and production began manually, so the human resources were not treated equally and just considered as the part to pay for work done where only basic knowledge about the job was sufficient.

In Industry 2.0, mass production began due to the invention of electricity. The industrialists realized the need for training and skilled employees as part of the HRM function. Along with basic skills, educational and technical skills were now important competencies (Hajimahmud et al., 2022).

In Industry 3.0, computerization and automation were replaced by machinery in which employees were considered to be the important assets of the organization. Engagement, motivation, and development functions of employees were prioritized and the need for managerial and human skills was raised and emerged.

Change is inevitable, so organizations and industries move with the pace of technology. Industry 4.0 emerged with AI, data science, and IoT, which enabled HR professionals to emphasize advanced training with specific competencies, such as innovation, creativity, design thinking, leadership, problem-solving skills, etc. so that individuals can meet and achieve organizational goals and objectives as shown in [Table 2.2](#).

TABLE 2.1
Evolution of Workforce Competencies with the Industrial Revolution

Industry Revolution	Year	Industry Evolution	HR Evolution	Competencies
1.0	1760	Mechanization, water power, steam power	Personnel management, employees need identification	Basic skills
2.0	1840	Mass production, assembly line, electricity	Recruitment and selection, training needs, reward system	Educational Technical skills
3.0	1970	Computers and automation	Employee development, employee relations, employee engagement	Automated skills Managerial skills
4.0	2011	Cyber-physical systems, artificial intelligence, data science, IoT (Khang et al., 2022a,b)	Advanced training using IoT (Rani et al., 2021), machine learning, competency mapping, succession planning, and retention strategies.	Innovation and creativity Leadership skills Design thinking and decision- making

TABLE 2.2
Definitions of Competency and Competency Model

Sr. No.	Author and Year	Definition
1	Boyatzis, 2008	An underlying feature of a person that results in effective and/or exceptional performance in a job.
2	Spector, 1997	Competency is a confluence of an individual’s attitudes, capabilities, skills, and other traits.
3	UNIDO, 2002	A competency is a group of abilities, knowledge, and characteristics that enable someone to carry out a task or activity within a certain role or function.
4	Khang, Rani, and Shah, 2024	Competency is the practical application of a person’s abilities, and it includes the knowledge, abilities, and attitudes necessary for successful professional performance.
5	Sparrow, 1995	A competency model is a guideline established by an HR department that sets forth the precise skills, knowledge, and behavioral requirements that help a person to complete their work efficiently.
6	Khang, Rani, and Shah, 2024	A modern competency model is developed by the AIoCF organization through global software development and IT services within the software industry and business. The current version of the competency framework adopted knowledge, skills, and tools that are now available and ready for you to apply for your corporation.

Competencies are incredibly organized and role-specific, which means they could not apply to different industries, businesses, or work functions. Over the past 30 years, the competency-based approach to HR management has become essential.

Organizations struggle to recruit qualified workers in the current environment, where employment availability is at a critical point and there is a high demand for efficient low-cost workers.

2.3 RELATED WORK

Every professional activity and every job must be characterized by a competency, which is a behavioral job description (Fogg, 1999). A certain job typically requires a set of seven to nine total skills, which are represented in a competency model depending on the workplace and organizational environment.

The employee selection process, which is currently based primarily on evaluating the key competencies of candidates for offered positions, is undoubtedly one of the most important personnel activities in Small and medium-sized enterprises (SMEs), including those in the trading sector. Therefore, candidates for positions in the field of trading, especially university graduates with degrees in business or economics, should have the desired level of necessary practical abilities and skills instead of theoretical knowledge to succeed in the selection process.

Due to the rising number of job applicants, the process of choosing qualified candidates is currently getting more difficult and demanding. HR managers typically adopt a competency-based strategy to choose the best candidate because of the requirements for matching the needs of firms, notably trading ones, with specific employee competencies (Snehal, Babasaheb & Khang, 2023).

The competency-based approach entails using a competency model to specify the required framework of candidates' essential skills for particular occupations as the basis for choosing qualified candidates for a given post.

Rodriguez et al. (2002) state that the benefit of competency models is that they can be used to develop a whole-person assessment or holistic approach to looking at the competencies that a person already possesses and may still need to pick up to perform well in a particular industry or occupation. Professionals in workforce development or HR development can then successfully use the knowledge in a variety of applications with the workforce.

Goyal (2017) explored the application of AI to improve the HRM screening process. A computer can read language successfully thanks to a process called natural language processing, which transforms text into structured and understandable data.

Natural language generation is the reverse of natural language processing, allowing a computer to write a language and organize facts into text. These two technologies have a lot of potential for recruiting top personnel. Consequently, AI could profit from human behavior and its implications. However, to effectively incorporate the processing of natural language, AI is required. This is because talking with humans requires a thorough understanding of their written and spoken communication patterns.

According to Yuvaraj (2021), the knowledge and ability of human resources to carry out a specified activity are heavily stressed throughout the text. The management and development of competencies are seen as essential for boosting organizational competitiveness. The literature demonstrates that in addition to excellent production, general, management, and technical abilities are crucial for an individual's professional advancement.

Garrett (2007) infers that a competency model is a qualitative instrument that evaluates the talents and skills required to carry out a certain role within a position, field, or organization.

The specific abilities are necessary for each organization to accomplish its mission and goals. Experts suggest that the competency model has proved to be one of the best tools for organizations to understand the methods to opt for and realize the levels at which the number of competencies is needed for an organization to succeed in terms of competitive advantage.

Lee (2010) stated that not all of the created model's competencies are equally important, so it is necessary to condense them into a small group with an emphasis on the key skills of top achievers. Fogg (1999) suggests that competency mapping is a crucial organizational process that supports knowledge management and other organizational goals.

Jain (2013) concluded that the HR departments of the largest pharmaceutical companies examine the competency mapping of various individuals to accomplish the specific duties assigned to them efficiently. This also gives them the chance to gain the necessary set of skills to get ready for the next responsibilities in the organization.

Chouhan and Srivastava (2014) noted that it was important to create fundamental proficiency given that the HR industry has earned a place at the strategic table. The importance of establishing a solid basic capability has been emphasized. High intellectual capital gives a company a competitive edge in the IT sector.

As per Balaji and Vimala's (2012) research conclusion, employment skills are observed to have greater capability gaps, which can be filled by providing targeted training to improve organizational performance. Chouhan and Srivastava (2014) noted that it was important to create fundamental proficiency. It has emphasized the necessity to build fundamental skills.

Howie et al. (2016), in their research and exploration of the connection between managerial organizational effectiveness and competency, the results revealed that each of the abilities vis-à-vis the ability to lead, solve problems, think strategically, and prioritize customers had a favorable impact on the organization's effectiveness.

Jatoba et al. (2019) in their study entitled "Evolution of AI in HR" extend theoretical research on the development of AI in the field of HR. There are many different motivations, goals, strategies, AI methodologies (Bhambri et al., 2022), and HR applications. It is claimed that the integration of AI and HR has not yet led to a theoretical breakthrough or the emergence of a new conceptual domain.

As a result, it can be inferred that enlisting the help of technology researchers and HR experts, as well as forming multidisciplinary teams made up of engineers and HR specialists, can reinforce this structure, as shown in [Figure 2.1](#).

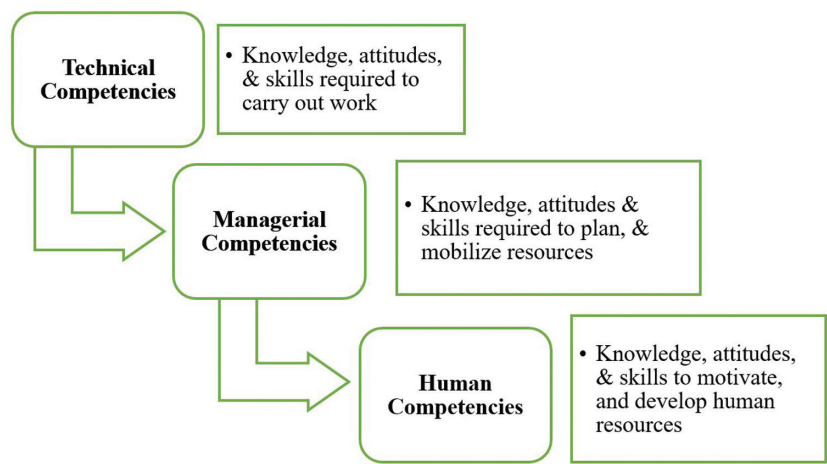


FIGURE 2.1 Classification of competencies.

2.4 COMPETENCY MODEL

The competency model includes the needed competencies for work, as well as the description of individual capabilities and performance and result indicators. These lists could have various levels of depth and could also explain how the competencies relate to each other.

Over the years, many competency models have been created. By dividing competencies into meta, domain, method, and social competencies, Competency models for management and leadership are provided by other writers. Additionally, competency models are available for specific jobs or tasks. According to Tucker and Cofsky (1994), competency has five main components, as shown in [Figure 2.2](#).

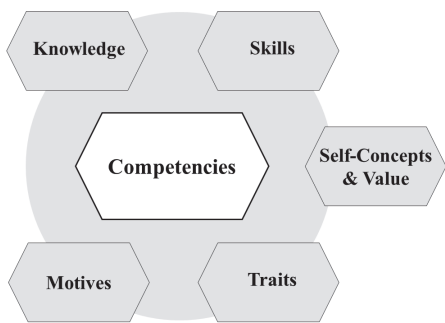


FIGURE 2.2 Competency model.

2.4.1 TUCKER AND COFSKY COMPETENCY MODEL

1. **Knowledge.** This is the understanding and learning that a person possesses.
2. **Skill:** This is the ability of a person to carry out a specific task.
3. **Self-Concepts and Values:** A person's attitudes, principles, and sense of self are covered in this category. A person's self-confidence, or their conviction that they can succeed in a particular circumstance, such as a surgical condition, is an example.
4. **Traits:** Physical qualities and dependable reactions to stimuli or information are referred to as traits. Surgeons must have good eyesight, self-control, and the capacity to remain composed under pressure.
5. **Motives** - Motives are feelings, wants, bodily demands, or similar urges that cause one to take action. For instance, surgeons who value interpersonal relationships assume personal accountability for getting along well with other personnel in the operating group.

The competency-based strategy entails using a competency model to specify the required framework of applicants' essential skills for certain occupations as the foundation for choosing qualified candidates for a post (Calvert, 2014; Campion, 2011).

2.4.2 AI-ORIENTED COMPETENCY FRAMEWORK

In first year of Industrial Revolution 4.0, according to the AI-oriented competency framework (AIoCF) designed by SCEDEX organization in 2011 is available for role competency assessments for corporations in the world.

In businesses, an effective competency model (AIoCF) includes several components. A new competency version 6.0 is the first version used for the human capital management system (HCMS); it is applied in information technology (IT) companies all over the world, and it has four key components as shown in [Figure 2.3](#).

1. **Knowledge:** Knowledge and education are the understanding and learning that a person possesses. Knowledge and education can be defined as awareness of facts or as practical skills, and may also refer to familiarity with objects or situations. Knowledge and education of facts, also called propositional knowledge, is often defined as true belief that is distinct from opinion or guesswork by virtue of justification.
2. **Skills:** Skills are the hard and soft skills that a person has. A skill is the learned ability to act with determined results and good execution, often within a given amount of time, energy, or both. Skills can often be divided into domain-general and domain-specific skills.
3. **Experiences:** Work experiences are the work and practice experience in which a person completes his or her roles and responsibilities. Employees are held accountable for completing several tasks in the workplace.
 - Role refers to one's position on a team. A role is a set of connected behaviors, rights, obligations, beliefs, and norms as conceptualized by people in a workplace. It is an expected, free, or continuously changing behavior and may give an individual social status or social position.

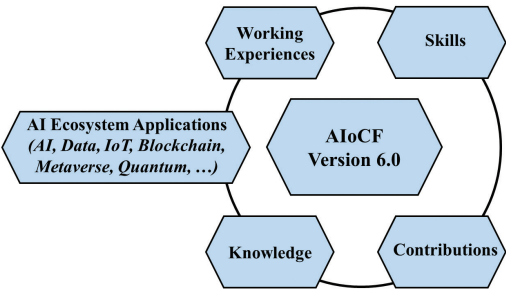


FIGURE 2.3 An AI-oriented competency model (AIoCF) designed by SCEDEX organization, 2011.

Source: SCEDEX organization.

- Responsibility refers to the tasks and duties of a particular role or job description that a person must complete in period of time.
4. **Contributions:** An employee’s contributions are the delivering and supporting that a person has used his/her own physical or virtual properties for supporting projects and colleagues in the company.

The outcome of the competency framework is used to build job descriptions and assessments for each job title or role in company.

In light of the increasing number of job applicants, the process of choosing qualified candidates is currently getting more difficult and demanding. HR managers typically adopt a competency-based strategy to choose the best candidate because of the requirements for matching the needs of firms, notably trading firms, with specific employee competencies.

2.4.3 EXAMPLES OF THE COMPETENCY MODEL

Here are some examples of the competency models proposed by a few authors.

2.4.3.1 The Capgemini Consulting Group

The Capgemini Consulting Group’s APM Maturity Model’s main objective is to assist businesses in managing performance, a crucial component of automation. With the help of the model, businesses may monitor the sustainability of their assets, enhance customer focus, and reduce the total cost of ownership. Cost savings and production plan management result from these advancements.

Determine the skills an organization needs to have to successfully adopt Industry 4.0 with the help of the Industry 4.0 MM from Uni Ankara. It serves as a roadmap for organizations to discover their flaws and the areas where they encounter issues. Best practices are also included that support their consistent transformation into Industry 4.0 businesses.



FIGURE 2.4 Core competencies of workforce.

Source: Adapted from Weber, Crawford, and Dennison (2012) and Rickman and Partridge (1998).

2.4.3.2 Weber, Crawford, & Rickman

In Weber, Crawford, and Dennison (2012), and Rickman and Partridge (1998), this interaction of the following core competency categories results in individual competency for desired or needed ideal behavior of an individual, as the illustration in [Figure 2.4](#) shows.

1. Problem-solving-related skills, such as the ability to negotiate, present, and process information.
2. Social abilities (relational, interpersonal): These include the capacity for communication, empathy, teamwork, the ability to inspire and lead others, the capacity for forming connections and relationships, etc.
3. Self-management skills, such as the capacity for independent work, adaptability, readiness to learn, self-motivation, and assertiveness.

2.4.3.3 AI-oriented Competency Framework 6.0

SCEDEX organization has defined competency frameworks with AI ecosystem application skills, problem-solving skills, social abilities skills, and self-management skills for the digital economy era as the illustration in [Figure 2.5](#) shows.

1. Problem-solving-related skills, such as the ability to negotiate, present, and process information.
2. Self-management skills, such as the capacity for independent work, adaptability, readiness to learn, self-motivation, and assertiveness.
3. Social abilities (relational, interpersonal): These include the capacity for communication, empathy, teamwork, the ability to inspire and lead others, the capacity for forming connections and relationships, etc.

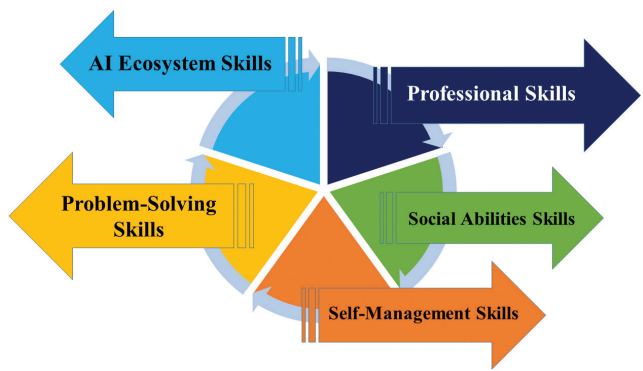


FIGURE 2.5 Core competencies of the workforce.

Source: Adapted from SCEDEX (2011).

- 4. AI ecosystem application skills: These include the knowledge, literacy, concepts, models, technologies, applications, and tools for communicating, guiding, learning or teaching, using or working in AI, IoT, blockchain, metaverse, and data science.

2.5 WORKFORCE COMPETENCIES

2.5.1 COMPETENCY FRAMEWORK FOR INDUSTRY 4.0

The success or failure of a competency model largely depends on its implementation and accurate application. We can say that it would be more successful if the model is implemented in every area of HRM, i.e., recruitment and selection, training and development, performance appraisal, compensation management, rewards and recognition, and retirement benefits of the employees. The employees in the organization are fully aware of the competencies necessary for success, as well as the evaluation criteria.

Based on the available related work, researchers are trying to figure out what skills future professionals will need to successfully adjust to the workforce, as shown in [Table 2.3](#).

2.5.2 COMPETENCY FRAMEWORK FOR THE DIGITAL ECONOMY

Along with popular competencies, the employees in the digital economy are fully aware of the competencies necessary for success as well as the evaluation criteria. Based on the available related work, researchers are trying to depict what skills future professionals will need to adjust to the workforce as shown in [Table 2.4](#).

TABLE 2.3

Competencies Necessary for the Workforce in Industry 4.0

Technical Competencies	Methodological Competencies	Social Competencies	Personal Competencies
State-of-the-art knowledge	Creativity	Intercultural skills	Flexibility
Technical skills	Entrepreneurial thinking	Language skills	Ambiguity tolerance
Process understanding	Problem-solving	Communication skills	Motivation to learn
Media skills	Conflict solving	Networking skills	Ability to work under pressure
Coding skills	Decision making	Ability to work in a team	Sustainability mindset
Understanding IT security (Khang et al., 2022a,b)	Ability to analyze	Ability to cooperate	Compliance
	Research skills	Ability to transfer knowledge	
	Efficiency	Leadership skills	

Source: Author’s owner.

TABLE 2.4

Competencies Necessary for the Workforce in the Digital Economy

Technical Competencies	Methodological Competencies	Social Competencies	Personal Competencies
State-of-the-art knowledge	Creativity	Intercultural skills	Flexibility
Technical skills	Entrepreneurial thinking	Language skills	Ambiguity tolerance
Process understanding	Problem-solving	Communication skills	Motivation to learn
Media skills	Conflict-solving	Networking skills	Ability to work under pressure
Coding skills	Decision-making	Ability to work in a team	Sustainability mindset
Understanding IT security	Ability to analyze	Ability to cooperate	Compliance
Understanding AI (machine learning, deep learning)	Ability to work	Ability to use or apply	Awareness
Understanding data science (data engineering, data analytics, big data)	Ability to work	Ability to use or apply	Awareness
Understanding IoT technology	Ability to work	Ability to use or apply	Awareness
Understanding blockchain technology	Ability to work	Ability to use or apply	Awareness
Understanding metaverse technology	Ability to work	Ability to use or apply	Awareness
	Research skills	Ability to transfer knowledge	
	Efficiency	Leadership skills	

Source: SCEDEX (2021).

2.6 REQUIRED COMPETENCIES AND KNOWLEDGE
ASSOCIATED WITH ELEMENTS OF INDUSTRY 4.0

According to a study conducted by industry officials and experts, the following competencies and knowledge are connected to various Industry 4.0 requirements. Transversal competencies include problem-solving skills, soft (personal) competencies, systems thinking, business thinking, and technological literacy, as shown in Table 2.4 above.

Transversal skills can be used in a variety of fields and are divided into three categories: primary, intermediate (based on basic), and high (based on intermediate). These skills are all connected and related to one another.

Competencies in problem-solving are essential for the Industry 4.0 strategy. These skills encompass a problem-solving mindset, fundamental sciences, and applied sciences.

Scientific methods and technological literacy aid in understanding and resolving problems. Through imaginative experimenting, cognitive processes might occasionally arise.

Leadership, networking, communication (oral and written), and assertiveness are examples of soft talents that enable people to work in multidisciplinary teams as shown in Table 2.5.

TABLE 2.5
Transversal or Cross-Sector Competencies

Problem-solving	Soft Abilities		Systems Analysis	Design Thinking	Technical Know-How
	Interpersonal	Metacognition and individual perspectives	Ethics and moral values		
Innovation and creativity	Communication	Self-confidence	Sustainable growth	Critical Thinking	Technical and software knowledge
Knowledge management	Teamwork	Responsibility	Societal change	Collaboration	
Scientific procedure	Leadership	Lifelong learning			
Knowledge administration	Assertiveness	Self-direction			
		Self-judgment			
		Self-management			
		Self-motivation			
		Self-knowledge			
		Self-regulation			
		Self-awareness			

Source: Author’s owner.

Industry 4.0 innovation depends on the advantages of know-how and the requirements of various interested parties. Analyzing the business aspect of goods and services is possible with the help of business thinking. Finally, technological literacy is defined as the understanding of fundamental engineering tools, how they work, and how they are applied (Rani et al., 2023).

People with managerial skills will also be in high demand in Industry 4.0. These are the people who manage teams and make business decisions. The industry has identified eight competencies as being crucial for managers. Every profession needs different competencies as a part of their work. Knowledge, experiences, aptitude, skills, contributions, ability, and attitude are common competencies, as shown in [Table 2.6](#).

2.7 REQUIRED KNOWLEDGE AND SKILLS ASSOCIATED WITH ELEMENTS OF THE DIGITAL ECONOMY

Basing on the model conducted by SCEDEX industry officials and experts, the following competencies and knowledge are connected to various requirements of AI ecosystem in the AI-oriented economy. Core competencies include professional skills, engineering skills, problem-solving skills, soft skills, and non-engineering skills. There are AI and data science skills in competency models that enable people to work in multidisciplinary teams as shown in [Table 2.7](#).

2.8 COMPETENCY FRAMEWORK AND DESIGN

Having an AIoCF can help organizations understand where you are concentrated, how prepared you are to meet future HR challenges, and how key employees can be developed and retained, while simultaneously improving the effectiveness of the organization in the era of digital economy.

AIoCF 6.0 has included five key components (knowledge, skills, experiences, AI ecosystem application, contributions) and their corresponding weight (%). They are dependent on the model of BUILD-UP and BUY-IN frameworks in the human capital development plan.

2.9 BUILD-UP COMPETENCY FRAMEWORK AND DESIGN

The BUILD-UP framework can be used to define job descriptions and job assessment for company employees in version 6.0 of the workforce development plan.

For BUILD-UP competency-based job descriptions, dependent on the specific business model and culture of the organization, you can add, update, and remove assessment items as well as set weight for each component in a competency framework. See an example of a software engineer role in [Figure 2.7](#).

2.10 BUY-IN COMPETENCY FRAMEWORK AND DESIGN

The BUY-IN framework can be used to define job descriptions and job assessments for newly hire employees in version 6.0 of the workforce development plan, as shown in [Figure 2.8](#).

TABLE 2.6**Essential Competencies for Various Domains of Industry 4.0**

Functions	Competency Dimensions	Domain-Specific Competencies				
		Engineers	Business	Healthcare	Design	IT
Leading and deciding	Initiating actions	Decision-making Taking initiative and responsibility				
	Leading and supervising	Leadership skills Possessing the traits essential to handle business				
Supporting and cooperating	Working with People	Teamwork Collaborating and communicating with people				
	Adhering to principles and values	Respecting ethics Environmental awareness Awareness of ergonomics				
Interacting and presenting	Relating and networking	Compromising Creating business networks Maintaining customer relationships				
	Persuading and influencing	Negotiating Emotional intelligence				
Analyzing and interpreting	Presenting and communicating information	Presentation and communication ability				
	Writing and reporting	Targeted/technical communication Literacy				
	Applying expertise and technology	IT and technology affinity Economics Extract business value from social media				

(Continued)

TABLE 2.6 (Continued)
Essential Competencies for Various Domains of Industry 4.0

Functions	Competency Dimensions	Domain-Specific Competencies				
		Engineers	Business	Healthcare	Design	IT
Creating and conceptualizing	Analyzing	Digital-to-physical technology transfer, network security, IT architectures, machine learning, integrating heterogeneous technologies, mobile technologies, sensors/embedded systems network, technology/M2M communication, robotics/AI, predictive maintenance, AR and VR (Khang et al., 2022a,b)	Tech-enabled processed: Forecasting, planning, scheduling, business analysis, digital skills, talent management and strategy, technology awareness, role of managers as facilitators	Structuring of healthcare systems and services, epidemiological analysis, healthcare principles and systems, integrity and ethical conduct, bioethics, evidence-based practice (Khang et al., 2023a,b)	Statistical data analysis and technique, simulation and modeling, understanding the impact of technology, tech-enabled product and service design, human-robot interaction, user interfaces	Service offerings, business process management, change management, workflow coordination, workflows, modeling, big data, data analysis and results, cloud computing, architectures, databases, statistics, data security
		Problem-solving				
		Optimization				
		Analytical skills				
		Cognitive ability				
	Learning and researching	Lifelong learning				
		Knowledge management				
		Innovating				
	Creating and innovating	Creativity				
		Critical thinking				
	Change management					
	Formulating strategies and concepts	Business strategy				
		Abstraction ability				
		Managing complexity				

(Continued)

TABLE 2.6 (Continued)
Essential Competencies for Various Domains of Industry 4.0

Functions	Competency Dimensions	Domain-Specific Competencies				
		Engineers	Business	Healthcare	Design	IT
Organizing and executing	Planning and organizing	Project management Planning and organizing work Management ability				
	Delivering results and meeting customer expectations	Customer orientation Customer relationship management				
	Following Instructions and Procedures	Legislation awareness Safety awareness Individual responsibility				
Adopting and coping	Adapting and responding to change	Work in interdisciplinary environments Intercultural competency Flexibility, adaptability, and ability to change mindset				
	Persuading and influencing	Work-life balance Handling work pressure Stress management				
	Achieving personal and work goals and objectives	Self-management and organization				
Enterprising and performing	Entrepreneurial and commercial thinking	Business model understanding Entrepreneurship				

Table 2.6 presents a part of their work knowledge, experiences, aptitude, skills, contributions, ability, and attitude.

TABLE 2.7
AI and Data Science Literacy and Skills

AI and Data Science Application Skills	IT Field	Non-IT Field	Managerial Field
AI science	To be applied	To be used	Awareness
Data science	To be applied	To be used	Awareness
IoT technology	To be applied	To be used	Awareness
Blockchain technology	To be applied	To be used	Awareness
Big data and analytics	To be applied	To be used	Awareness
Metaverse	To be applied	To be used	Awareness

Source: SCEDEX (2021).

For the BUY-IN competency-based job description, dependent on the specific recruitment model, you can add, update, and remove assessment items as well as set weight for each component in the competency framework, with an example of the software engineer role in [Figure 2.9](#).

2.11 RESULTS AND DISCUSSION

Once BUILD-UP and BUY-IN competency frameworks are designed, they are aggregated into a framework that can be used by HR department to define the requirements of the competency-based job descriptions for success in each specific role in businesses. These competency frameworks are the set of experiences, skills, knowledge, contributions, and others for specific roles as [Figure 2.6](#).

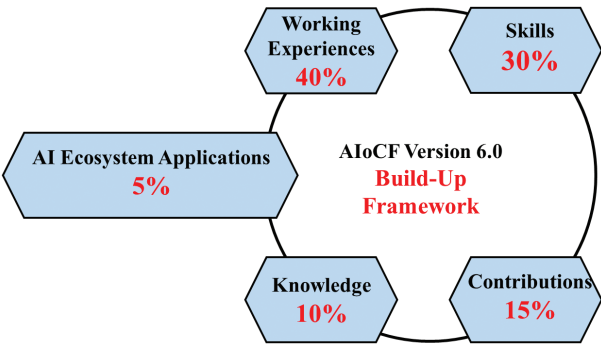


FIGURE 2.6 BUILD-UP competency framework designed by SCEDEX organization, 2011 (SCEDEX, 2011).

Software Engineer Role		BUILD-UP
Assessment Items	Code	Weight %
Roles & Responsibilities		100%
Roles and Responsibilities	AD01	10%
Project Diversity and Complexity	AD02	10%
Experiences & Contributions		20%
Software Engineering Experiences (*)	AD03	5%
Subordinate Development	AD04	5%
Number of Applications or Software Projects	AD05	4%
SME Community Contributions	AD06	3%
Business Industry Experiences	AD07	2%
Solutions Consulting for Sales/Bidding Teams	AD08	1%
Software Engineering Professional Skills		30%
Software Requirement Analysis	AD09	5%
Architecture Design and Software Designer	AD10	5%
Computer Programming Languages (*)	AD11	5%
Application Software Development and Services (*)	AD12	10%
Application Software Quality Inspection	AD13	3%
Web API and Microservices Development	AD14	2%
Software Engineering Optional Skills		10%
Storage and Database Development	AD15	3%
Cloud Platforms and Application Migration	AD16	3%
AI, IoT, Blockchain, Data Science Skills	AD17	3%
Version Control and DevOps Services	AD18	1%
Foreign Language & Certificates		10%
Foreign Language	AD19	5%
Education Background	AD20	2%
Software Engineering Certifications	AD21	2%
Individual Learning in Year	AD22	1%
Non-Engineering and Softskills		10%
Interpersonal Skills	AD23	3%
Scrum / Agile Model	AD24	2%
Troubleshooting and Technical Support	AD25	2%
Software Documentation and Guidelines	AD26	2%
Software Project Management	AD27	1%

FIGURE 2.7 A BUILD-UP competency-based job description designed by SCEDEX organization. An example of a software engineer role (SCEDEX, 2011).

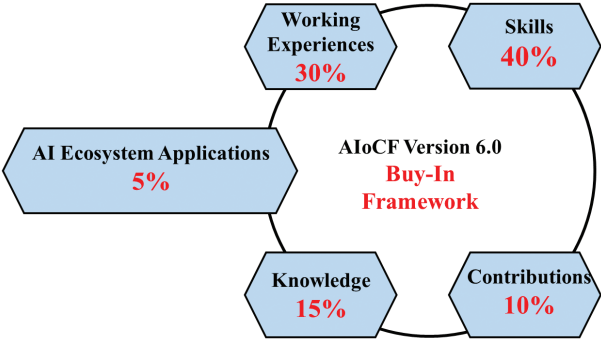


FIGURE 2.8 BUY-IN competency framework designed by SCEDEX organization, 2011.

In fact, competency frameworks are used for recruitment, hiring, learning, and development (L&D) department efforts as well as working performance assessments.

2.12 CONCLUSION

Industry 4.0’s primary objective is to increase the productivity and adaptability of factories so that they can meet future demands. A variety of technologies make it possible to use this strategy, such as robots (Tailor, Pareek & Khang, 2022), simulation, system integration, IoT, cybersecurity, the cloud, additive manufacturing, AR/VR, big data analytics, etc.

All of these enable businesses to connect and utilize the enormous amount of information produced for better decision-making. The application of Industry 4.0 has changed the function of a human resource manager from a manual administrative activity to one that involves more strategic thinking.

The competencies required for Industry 4.0 have been extensively covered in the literature. The most typical ones are categorized as knowledge and experience, Industry 4.0 technologies, data analysis, and technical expertise. However, there is no agreement on the skills required to be productive in Industry 4.0 contexts. The capacity to use their knowledge in various collaborative contexts to add value is a crucial skill for future professionals.

The workforce in the era of digital economy must learn from new environments and experiences. Businesses must take into account the requirement for employees to continually develop new skills (Khang, Rani, and Shah, 2024).

It should be obvious from the study that it is a complex process to build and apply competencies. The tried-and-true method of investing in the workforce to create a more efficient and effective staff is creating and implementing a competency model.

An organization’s ability to function depends on several exceptional factors, with the skilled person playing a key part in achieving organizational objectives. Due to

Software Engineer Role		BUILD-UP	BUY-IN
Assessment Items	Code	Weight %	Weight %
		100%	100%
Roles & Responsibilities		20%	0%
Roles and Responsibilities	AD01	10%	0%
Project Diversity and Complexity	AD02	10%	0%
Experiences & Contributions		20%	40%
Software Engineering Experiences (*)	AD03	5%	25%
Subordinate Development	AD04	5%	0%
Number of Applications or Software Projects	AD05	4%	5%
SME Community Contributions	AD06	3%	0%
Business Industry Experiences	AD07	2%	10%
Solutions Consulting for Sales/Bidding Teams	AD08	1%	0%
Software Engineering Professional Skills		30%	30%
Software Requirement Analysis	AD09	5%	5%
Architecture Design and Software Designer	AD10	5%	5%
Computer Programming Languages (*)	AD11	5%	5%
Application Software Development and Services (*)	AD12	10%	10%
Application Software Quality Inspection	AD13	3%	3%
Web API and Microservices Development	AD14	2%	2%
Software Engineering Optional Skills		10%	10%
Storage and Database Development	AD15	3%	3%
Cloud Platforms and Application Migration	AD16	3%	3%
AI, IoT, Blockchain, Data Science Skills	AD17	3%	3%
Version Control and DevOps Services	AD18	1%	1%
Foreign Language & Certificates		10%	10%
Foreign Language	AD19	5%	5%
Education Background	AD20	2%	2%
Software Engineering Certifications	AD21	2%	2%
Individual Learning in Year	AD22	1%	1%
Non-Engineering and Softskills		10%	10%
Interpersonal Skills	AD23	3%	3%
Scrum / Agile Model	AD24	2%	2%
Troubleshooting and Technical Support	AD25	2%	2%
Software Documentation and Guidelines	AD26	2%	2%
Software Project Management	AD27	1%	1%

FIGURE 2.9 BUY-IN competency-based job description designed by SCEDEX organization, with the example of a software engineer role.

the accelerating rate of change in the corporate environment, competency modeling in the current economic climate requires a forward-looking and proactive strategy (Hussain, Sivakumar & Khang, 2022).

Competencies can be used to translate strategy into individual and work-related skills. Individuals can manage their career success, navigate their career path, or apply knowledge to examine new opportunities (Khang et al., 2023a,b).

We hope this chapter will help you understand the competency frameworks in the workforce development model. If you're looking to build a high-quality, flexible, and

enriching job description and job assessment documents, you'll need to know the whole competency structure.

Actually, there are many benefits of competency modeling. Although combination of these models for each role at your company might at first seem repetitive, there will be a fair amount of overlap across roles. It's hard to figure out where to start, and of course this section can lend you a helping hand! Explore more presentations and competency-based job descriptions in the workforce development model and other human capital management platforms in the book Khang (2023), *AI-oriented Competency Framework for Talent Management in the Digital Economy: Models, Technologies, Applications, and Implementation* (CRC Press).

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3 AI Powered Workforce Management in Industry 4.0 Era

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3.1 INTRODUCTION

Computer technology know-how defines artificial intelligence (AI) studies because they have a look at of “sensible agents”: any tool that perceives its surroundings and takes movements that maximize its risk of efficiently reaching its desires.

Kaplan and Haenlein (2018) describe AI as “a system’s capacity to successfully interpret outside data, to research from such data, and to apply the ones learning to obtain particular desires and obligations via bendy adaptation” (Khang et al., 2023a).

AI and machine learning use algorithms that offer answers for complicated issues and show sensible conduct akin to any human intelligence.